Matthias **Mittner**

Professor of Cognitive Neuroscience
University of Tromsø, Department of Psychology, 9037 Tromsø, Norway
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I am a professor at the UiT - The arctic university of Norway in Tromsø and at the Norwegian University of Science and Technology in Trondheim. My main interests within the field of cognitive neuroscience are mind wandering, non-invasive brain stimulation, decision making, pupillometry and selective attention. I am also interested in the development of methodological approaches in my field and enjoy computational methods, cognitive modeling, machine-learning and Bayesian statistics.

Education

Dr. phil. — **University of Frankfurt, Germany**
Psychology — 21.12.2011
- Thesis: 'Binding and Retrieval of Stimulus-Response Episodes in the Negative Priming Paradigm'
- Grade: Summa cum laude

B. Sc. — **Institute of Computer Science, University of Göttingen, Germany**
Computer Science — 2005/07 - 2008/04
- Thesis: 'Single Trial Estimation and Timewarped Averaging of Event-Related Potentials'

Diploma — **Department of Psychology, University of Göttingen, Germany**
Psychology — 2005/07 - 2007/11
- Thesis: 'Negative Priming and Response-Relation: Behavioural and Electroencephalographic Correlates'

Student at the University of Oslo — **Oslo, Norway**
Psychology and Computer Science — 2004/08 - 2005/06
- Erasmus-Scholarship; studies of computer science and psychology

B. Sc. (equivalent) — **Department of Psychology, University of Göttingen, Germany**
Psychology — 2002/09 - 2004/07
- Thesis: 'An Adaptive Approach to Memory-Span Tasks'

High-School Diploma — **Alexander von Humboldt Gymnasium, Eberswalde, Germany**
Main subjects physics and mathematics — 2002/06

Employments

Professor II — **Trondheim, Norway**
Quantitative Methods — 2021/03 - present
- Department of Psychology, Norwegian University of Science and Technology, Trondheim

Professor — **Tromsø, Norway**
Cognitive Neuroscience — 2020/09 - present
- Institute of Psychology, University of Tromsø, Tromsø

Associate Professor — **Tromsø, Norway**
Cognitive Neuroscience — 2014/04 - 2020/09
- Institute of Psychology, University of Tromsø, Tromsø

Post-Doctoral Researcher — **Amsterdam, The Netherlands**
Cognitive Neuroscience — 2013/04 - 03/2014
- Cognitive Science Center Amsterdam, Universiteit van Amsterdam
- Supervisor: Prof. Birte Forstmann

Post-Doctoral Researcher — **Göttingen, Germany**
Theoretical Neuroscience — 2012/09 - 2013/03

- Max-Planck Institute for Experimental Medicine, Göttingen
- Supervisor: Prof. Robert Gütig

PhD-Student — Göttingen, Germany

Computational Neuroscience — 2008/10 - 2012/01

- Max-Planck Institute for Dynamics and Self-Organization, Göttingen
- Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)
- Doctoral Program: Theoretical and Computational Neuroscience (PTCN)
- Supervisor: Prof. Marcus Hasselhorn

Guest Researcher — Göttingen, Germany

Computational Neuroscience — 2008/03 - 2008/09

- Max-Planck Institute for Dynamics and Self-Organization, Göttingen
- Project: cognitive modelling and data-mining (EEG)

Guest Researcher — Saarbrücken, Germany

Computer graphics — 2007/11 - 2008/02

- Max-Planck Institute for Informatics, Saarbrücken
- Project: optical modelling and psychophysics

Research Assistant — Göttingen, Germany

Computational Neuroscience — 2005/09 - 2007/10

- Bernstein Center for Computational Neuroscience, Göttingen
- Project: cognitive modelling; psychophysical experiments

Student Assistant — Göttingen, Germany

Psychology — 2003/09 - 2004/08

- Department for Developmental Psychology, University of Göttingen
- Duties: computer administration and development of experimental software

Publications

Books

(76) Mehmetoglu, M. & *Mittner, M.* (2021). *Applied Statistics Using R*. SAGE. https://tinyurl.com/svddmm

(75) Mehmetoglu, M. & *Mittner, M.* (2020). *Innføring i R for statistiske dataanalyser*. Universitetsforlaget. https://tinyurl.com/wx4xbfpc

Preprints

(74) Aasen, S. R., Babiker, S. M. E., Vangen, A. M. K., Haugan, S. B., Sandell, S. A. O., Saksvik-Lehouillier, I., Kam, J., *Mittner, M.*, & Csifcsák, G. (2025). The Effects of Partial Sleep Deprivation on Mind Wandering, Executive Performance and Affect. *Preprint*. https://osf.io/kmqfz_v3/ (submitted)

Journal articles

(73) Csifcsák, G. & *Mittner, M.* (2026). Reward Processing under Illusion of Control: The Sensitivity of the Feedback-Related Negativity to Prediction Errors Is Not Altered When Random Outcomes Are Perceived as the Consequence of One's Own Actions. *International Journal of Psychophysiology*, *219*, 113300. https://doi.org/10.1016/j.ijpsycho.2025.113300

(72) Alexandersen, A., Aasen, S. R., Csifcsák, G., & *Mittner, M.* (2025). Evidence for the Efficacy of Theta-Burst Stimulation above the Angular Gyrus for Reducing Mind Wandering Propensity: A Successful Pre-Registered Study. *Cortex*. https://doi.org/10.1016/j.cortex.2025.07.006

(71) Alexandersen, A., Dahlberg, K., Csifcsák, G., & *Mittner, M.* (2025). Unravelling the Threads of Thought: Probing the Impact of Contextual Factors on Mind Wandering. *Consciousness and Cognition*, *131*, 103870. https://doi.org/10.1016/j.concog.2025.103870

(70) Babiker, S., Luzzi, F., *Mittner, M.*, & Csifcsák, G. (2025). The Interplay Between Experimental Heat Pain and Noninvasive Stimulation of the Medial Prefrontal Cortex on Reinforcement Learning With Manipulated Outcome Contingencies. *European Journal of Neuroscience*, *61*(7), e70089. https://doi.org/10.1111/ejn.70089

(69) Drevland, R. N., Aasen, S. R., Csifcsák, G., & *Mittner, M.* (2025). Reducing Mind Wandering Using Continuous Theta Burst Stimulation. *Transcranial Magnetic Stimulation*, 100181. https://doi.org/10.1016/j.transm.2025.100181

(68) Mehmetoglu, M., Määttänen, I., & *Mittner, M.* (2025). The Link Between Mate Value Discrepancy and Relationship Satisfaction—An Empirical Examination Using Response Surface Analysis. *Behavioral Sciences*. https://doi.org/10.3390/bs15081131

(67) *Mittner, M.* & Groot, J. M. (2025). A Novel Method for Modeling Tonic and Phasic Pupil Dynamics in Humans. *Behavior Research Methods*, *57*(9), 233. https://doi.org/10.3758/s13428-025-02755-7

(66) Schultheiss, D. L., Turi, Z., Marmavula, S., Reinacher, P. C., Demerath, T., Straehle, J., Boedecker, J., *Mittner, M.*, & Vlachos, A. (2025). Efficient Prospective Electric Field-Informed Localization of Motor Cortical Targets of Transcranial Magnetic Stimulation. *Imaging Neuroscience*. https://doi.org/10.1162/IMAG.a.1056

(65) Vékony, T., Farkas, B. C., Brezóczki, B., *Mittner, M.*, Csifcsák, G., Simor, P., & Németh, D. (2025). Mind Wandering Enhances Statistical Learning. *iScience*, *28*(2). https://doi.org/10.1016/j.isci.2024.111703

(64) Aasen, S. R., Drevland, R. N., Csifcsák, G., & *Mittner, M.* (2024). Increasing Mind Wandering with Accelerated Intermittent Theta Burst Stimulation over the Left Dorsolateral Prefrontal Cortex. *Neuropsychologia*, *204*, 109008. https://doi.org/10.1016/j.neuropsychologia.2024.109008

(63) Groot, J. M., Miletic, S., Isherwood, S. J., Tse, D. H. Y., Habli, S., Håberg, A. K., Bazin, P., *Mittner, M.*, & Forstmann, B. U. (2024). A High-Resolution 7 Tesla Resting-State fMRI Dataset Optimized for Studying the Subcortex. *Data in Brief*.

(62) Mehmetoglu, M., Määttänen, I., & *Mittner, M.* (2024). Testing Sexual Strategy Theory in Norway. *Behavioral Sciences*, *14*(6), 438. https://doi.org/10.3390/bs14060438

(61) Sluppaug, K. S., Mehmetoglu, M., & *Mittner, M.* (2024). Modsem: An R Package for Estimating Latent Interactions and Quadratic Effects. *Structural Equation Modeling: A Multidisciplinary Journal*, *0*(0), 1-13. https://doi.org/10.1080/10705511.2024.2417409

(60) Groot, J. M., Miletic, S., Isherwood, S. J., Tse, D. H., Habli, S., Håberg, A. K., Forstmann, B. U., Bazin, P., & *Mittner, M.* (2023). Echoes from Intrinsic Connectivity Networks in the Subcortex. *Journal of Neuroscience*. https://doi.org/10.1523/JNEUROSCI.1020-23.2023

(59) Kreis, I., Zhang, L., *Mittner, M.*, Syla, L., Lamm, C., & Pfuhl, G. (2023). Aberrant Uncertainty Processing Is Linked to Psychotic-like Experiences, Autistic Traits, and Is Reflected in Pupil Dilation during Probabilistic Learning. *Cognitive, Affective, & Behavioral Neuroscience*. https://doi.org/10.3758/s13415-023-01088-2

(58) Nawani, H., *Mittner, M.*, & Csifcsák, G. (2023). Modulation of Mind Wandering Using Transcranial Direct Current Stimulation: A Meta-Analysis Based on Electric Field Modeling. *NeuroImage*, 120051. https://doi.org/10.1016/j.neuroimage.2023.120051

(57) Ørbo, M. C., Grønli, O. K., Larsen, C., Vangberg, T. R., Friborg, O., Turi, Z., *Mittner, M.*, Csifcsak, G., & Aslaksen, P. M. (2023). The Antidepressant Effect of Intermittent Theta Burst Stimulation (iTBS): Study Protocol for a Randomized Double-Blind Sham-Controlled Trial. *Trials*, *24*(1), 627. https://doi.org/10.1186/s13063-023-07674-6

(56) Rasmussen, I. D., *Mittner, M.*, Boayue, N. M., Csifcsák, G., & Aslaksen, P. M. (2023). Tracking the Current in the Alzheimer's Brain - Systematic Differences between Patients and Healthy Controls in the Electric Field Induced by tDCS. *Neuroimage: Reports*, *3*(2), 100172. https://doi.org/10.1016/j.ynirp.2023.100172

(55) Sedlinska, T., Bolte, L., Melsæter, E., *Mittner, M.*, & Csifcsák, G. (2023). Transcranial Direct-Current Stimulation Enhances Pavlovian Tendencies during Intermittent Loss of Control. *Frontiers in Psychiatry*, *14*. https://doi.org/10.3389/fpsyt.2023.1164208

(54) Alexandersen, A., Csifcsák, G., Groot, J., & *Mittner, M.* (2022). The Effect of Transcranial Direct Current Stimulation on the Interplay between Executive Control, Behavioral Variability and Mind Wandering: A Registered Report. *Neuroimage: Reports*, *2*(3), 100109. https://doi.org/10.1016/j.ynirp.2022.100109

(53) Groot, J. M., Csifcsák, G., Wientjes, S., Forstmann, B. U., & *Mittner, M.* (2022). Catching Wandering Minds with Tapping Fingers: Neural and Behavioral Insights into Task-unrelated Cognition. *Cerebral Cortex*, bhab494. https://doi.org/10.1093/cercor/bhab494

(52) Hawkins, G. E., *Mittner, M.*, Forstmann, B. U., & Heathcote, A. (2022). Self-Reported Mind Wandering Reflects Executive Control and Selective Attention. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-022-02110-3

(51) Kam, J. W. Y., *Mittner, M.*, & Knight, R. T. (2022). Mind-Wandering: Mechanistic Insights from Lesion, tDCS, and iEEG. *Trends in Cognitive Sciences*, *0*(0). https://doi.org/10.1016/j.tics.2021.12.005

(50) Csifcsák, G., Bjørkøy, J., Kuyateh, S., Reithe, H., & *Mittner, M.* (2021). Transcranial Direct Current Stimulation above the Medial Prefrontal Cortex Facilitates Decision-Making Following Periods of Low Outcome Controllability. *eNeuro*. https://doi.org/10.1523/ENEURO.0041-21.2021

(49) Groot, J. M., Boayue, N. M., Csifcsák, G., Boekel, W., Huster, R., Forstmann, B. U., & *Mittner, M.* (2021). Probing the Neural Signature of Mind Wandering with Simultaneous fMRI-EEG and Pupillometry. *NeuroImage*, *224*, 117412. https://doi.org/10.1016/j.neuroimage.2020.117412

(48) Kreis, I., Biegler, R., Tjelmeland, H., *Mittner, M.*, Reitan, S. K., & Pfuhl, G. (2021). Overestimation of Volatility in Schizophrenia and Autism? A Comparative Study Using a Probabilistic Reasoning Task. *PLOS ONE*, *16*(1), e0244975. https://doi.org/10.1371/journal.pone.0244975

(47) Rasmussen, I. D., Boayue, N. M., *Mittner, M.*, Bystad, M., Grønli, O. K., Vangberg, T. R., Csifcsák, G., & Aslaksen, P. M. (2021). High-Definition Transcranial Direct Current Stimulation Improves Delayed Memory in Alzheimer's Disease Patients: A Pilot Study Using Computational Modeling to Optimize Electrode Position. *Journal of Alzheimer's Disease*, *83*(2). https://doi.org/10.3233/JAD-210378

(46) Turi, Z., Lenz, M., Paulus, W., *Mittner, M.*, & Vlachos, A. (2021). Selecting Stimulation Intensity in Repetitive Transcranial Magnetic Stimulation Studies: A Systematic Review between 1991 and 2020. *European Journal of Neuroscience*, *n/a*(n/a). https://doi.org/10.1111/ejn.15195

(45) Zmeykina, E., *Mittner, M.*, Paulus, W., & Turi, Z. (2021). Short-Lived Alpha Power Suppression Induced by Low-intensity Arrhythmic rTMS. *Neuroscience*, *466*, 1-9. https://doi.org/10.1016/j.neuroscience.2021.04.027

(44) Boayue, N. M., Csifcsák, G., Kreis, I. V., Schmidt, C., Finn, I., Vollsund, A. E. H., & *Mittner, M.* (2020). The Interplay between Executive Control, Behavioral Variability and Mind Wandering: Insights from a High-Definition Transcranial Direct-Current Stimulation Study. *European Journal of Neuroscience*, *n/a*(n/a). https://doi.org/10.1111/ejn.15049

(43) *Mittner, M.* (2020). Pypillometry: A Python Package for Pupillometric Analyses. *Journal of Open Source Software*, *5*(51), 2348. https://doi.org/10.21105/joss.02348

(42) Turi, Z., *Mittner, M.*, Lehr, A., Bürger, H., Antal, A., & Paulus, W. (2020). Theta-Gamma Cross-Frequency Transcranial Alternating Current Stimulation over the Trough Impairs Cognitive Control. *eNeuro*. https://doi.org/10.1523/ENEURO.0126-20.2020

(41) Zmeykina, E., *Mittner, M.*, Paulus, W., & Turi, Z. (2020). Weak rTMS-induced Electric Fields Produce Neural Entrainment in Humans. *Scientific Reports*, *10*(1), 11994. https://doi.org/10.1038/s41598-020-68687-8

(40) Boayue, N. M., Csifcsák, G., Aslaksen, P., Turi, Z., Antal, A., Groot, J., Hawkins, G. E., Forstmann, B., Opitz, A., Thielscher, A., & *Mittner, M.* (2019). Increasing Propensity to Mind-Wander by Transcranial Direct Current Stimulation? A Registered Report. *The European Journal of Neuroscience*. https://doi.org/10.1111/ejn.14347

(39) Csifcsák, G., Boayue, N. M., Aslaksen, P. M., Turi, Z., Antal, A., Groot, J., Hawkins, G. E., Forstmann, B. U., Opitz, A., Thielscher, A., & *Mittner, M.* (2019). Commentary: "Transcranial Stimulation of the Frontal Lobes Increases Propensity of Mind-Wandering without Changing Meta-Awareness". *Frontiers in Psychology*, *10*. https://doi.org/10.3389/fpsyg.2019.00130

(38) Csifcsák, G., Melsæter, E., & *Mittner, M.* (2019). Intermittent Absence of Control during Reinforcement Learning Interferes with Pavlovian Bias in Action Selection. *Journal of Cognitive Neuroscience*, 1-18. https://doi.org/10.1162/jocn_a_01515

(37) Hawkins, G. E., *Mittner, M.*, Forstmann, B. U., & Heathcote, A. (2019). Modeling Distracted Performance. *Cognitive Psychology*, *112*, 48-80. https://doi.org/10.1016/j.cogpsych.2019.05.002

(36) Hetland, A., Kjelstrup, E., *Mittner, M.*, & Vitterso, J. (2019). The Thrill of Speedy Descents: A Pilot Study on Differences in Facially Expressed Online Emotions and Retrospective Measures of Emotions During a Downhill Mountain-Bike Descent. *Frontiers in Psychology*, *10*. https://doi.org/10.3389/fpsyg.2019.00566

(35) Turi, Z., Csifcsák, G., Boayue, N. M., Aslaksen, P., Antal, A., Paulus, W., Groot, J., Hawkins, G. E., Forstmann, B., Opitz, A., Thielscher, A., & *Mittner, M.* (2019). Blinding Is Compromised for Transcranial Direct Current Stimulation at 1 mA for 20 Min in Young Healthy Adults. *European Journal of Neuroscience*, *0*(0). https://doi.org/10.1111/ejn.14403

(34) Boayue, N. M., Csifcsák, G., Puonti, O., Thielscher, A., & *Mittner, M.* (2018). Head Models of Healthy and Depressed Adults for Simulating the Electric Fields of Non-Invasive Electric Brain Stimulation. *F1000Research*, *7*, 704. https://doi.org/10.12688/f1000research.15125.2

(33) Csifcsák, G., Boayue, N. M., Puonti, O., Thielscher, A., & *Mittner, M.* (2018). Effects of Transcranial Direct Current Stimulation for Treating Depression: A Modeling Study. *Journal of Affective Disorders*, *234*, 164-173. https://doi.org/10.1016/j.jad.2018.02.077

(32) Hetland, A., Vittersø, J., Oscar Bø Wie, S., Kjelstrup, E., *Mittner, M.*, & Dahl, T. I. (2018). Skiing and Thinking About It: Moment-to-Moment and Retrospective Analysis of Emotions in an Extreme Sport. *Frontiers in Psychology*, *9*(971). https://doi.org/10.3389/fpsyg.2018.00971

(31) Turi, Z., Schäfer, S. A., Antal, A., Paulus, W., & *Mittner, M.* (2018). Data from 'Placebo Enhances Reward Learning in Healthy Individuals'. *Journal of Open Psychology Data*, *6*(1), 2. https://doi.org/10.5334/jopd.34

(30) Turi, Z., Bjørkedal, E., Gunkel, L., Antal, A., Paulus, W., & *Mittner, M.* (2018). Evidence for Cognitive Placebo and Nocebo Effects in Healthy Individuals. *Scientific Reports*, *8*(1), 17443. https://doi.org/10.1038/s41598-018-35124-w

(29) Csifcsák, G. & *Mittner, M.* (2017). Linking Brain Networks and Behavioral Variability to Different Types of Mind-Wandering. *Proceedings of the National Academy of Sciences of the United States of America*, *114*(30). https://doi.org/10.1073/pnas.1705108114

(28) Hawkins, G. E., *Mittner, M.*, Forstmann, B. U., & Heathcote, A. (2017). On the Efficiency of Neurally-Informed Cognitive Models to Identify Latent Cognitive States. *Journal of Mathematical Psychology*, *76*, 142-155. https://doi.org/10.1016/j.jmp.2016.06.007

(27) Turi, Z., *Mittner, M.*, Paulus, W., & Antal, A. (2017). Placebo Intervention Enhances Reward Learning in Healthy Individuals. *Scientific Reports*, *7*, 41028. https://doi.org/10.1038/srep41028

(26) *Mittner, M.*, Hawkins, G. E., Boekel, W., & Forstmann, B. U. (2016). A Neural Model of Mind Wandering. *Trends in Cognitive Sciences*, *20*(8), 570-578. https://doi.org/10.1016/j.tics.2016.06.004

(25) Rodríguez-Aranda, C., *Mittner, M.*, & Vasylenko, O. (2016). Association Between Executive Functions, Working Memory, and Manual Dexterity in Young and Healthy Older Adults: An Exploratory Study. *Perceptual and Motor Skills*, *122*(1), 165-192. https://doi.org/10.1177/0031512516628370

(24) Hawkins, G. E., Mittner, M., Boekel, W., Heathcote, A., & Forstmann, B. U. (2015). Toward a Model-Based Cognitive Neuroscience of Mind Wandering. *Neuroscience*, *310*, 290-305. https://doi.org/10.1016/j.neuroscience.2015.09.053

(23) Høifødt, R. S., *Mittner, M.*, Lillevoll, K., Katla, S. K., Kolstrup, N., Eisemann, M., Friborg, O., & Waterloo, K. (2015). Predictors of Response to Web-Based Cognitive Behavioral Therapy With High-Intensity Face-to-Face Therapist Guidance for Depression: A Bayesian Analysis. *Journal of Medical Internet Research*, *17*(9), e197. https://doi.org/10.2196/jmir.4351

(22) Turi, Z., *Mittner, M.*, Opitz, A., Popkes, M., Paulus, W., & Antal, A. (2015). Transcranial Direct Current Stimulation over the Left Prefrontal Cortex Increases Randomness of Choice in Instrumental Learning. *Cortex*, *63*, 145-154. https://doi.org/10.1016/j.cortex.2014.08.026

(21) *Mittner, M.*, Behrendt, J., Menge, U., Titz, C., & Hasselhorn, M. (2014). Response-Retrieval in Identity Negative Priming Is Modulated by Temporal Discriminability. *Frontiers in Psychology*, *5*, 621. https://doi.org/10.3389/fpsyg.2014.00621

(20) *Mittner, M.*, Boekel, W., Tucker, A. M., Turner, B. M., Heathcote, A., & Forstmann, B. U. (2014). When the Brain Takes a Break: A Model-Based Analysis of Mind Wandering. *Journal of Neuroscience*, *34*(49), 16286-16295. https://doi.org/10.1523/JNEUROSCI.2062-14.2014

(19) Ihrke, M., Behrendt, J., Schrobsdorff, H., Visser, I., & Hasselhorn, M. (2013). Negative Priming Persists in the Absence of Response-Retrieval. *Experimental Psychology*, *60*(1), 12-21. https://doi.org/10.1027/1618-3169/a000169

(18) *Mittner, M.* (2013). Functional Integration of Large-Scale Brain Networks. *Journal of Neuroscience*, *33*(48), 18710-18711. https://doi.org/10.1523/JNEUROSCI.4084-13.2013

(17) Schrobsdorff, H., Ihrke, M., Behrendt, J., Herrmann, J. M., & Hasselhorn, M. (2012). Identity Negative Priming: A Phenomenon of Perception, Recognition or Selection?. *PloS One*, *7*(3), e32946. https://doi.org/10.1371/journal.pone.0032946

(16) Schrobsdorff, H., Ihrke, M., Behrendt, J., Hasselhorn, M., & Herrmann, J. M. (2012). Inhibition in the Dynamics of Selective Attention: An Integrative Model for Negative Priming. *Frontiers in Psychology*, *3*, 491. https://doi.org/10.3389/fpsyg.2012.00491

(15) Ihrke, M. & Behrendt, J. (2011). Automatic Generation of Randomized Trial Sequences for Priming Experiments. *Frontiers in Psychology*, *2*, 225. https://doi.org/10.3389/fpsyg.2011.00225

(14) Ihrke, M., Schrobsdorff, H., & Herrmann, J. M. (2011). Recurrence-Based Estimation of Time-Distortion Functions for ERP Waveform Reconstruction. *International Journal of Neural Systems*, *21*(1), 65-78. https://doi.org/10.1142/S0129065711002651

(13) Ihrke, M., Behrendt, J., Schrobsdorff, H., Michael Herrmann, J., & Hasselhorn, M. (2011). Response-Retrieval and Negative Priming: Encoding- and Retrieval-Specific Effects.. *Experimental Psychology*, *58*(2), 154-161. https://doi.org/10.1027/1618-3169/a000081

(12) Ihrke, M. & Brennen, T. (2011). Sharing One Biographical Detail Elicits Priming between Famous Names: Empirical and Computational Approaches. *Frontiers in Psychology*, *2*(75). https://doi.org/10.3389/fpsyg.2011.00075

(11) Behrendt, J., Gibbons, H., Schrobsdorff, H., Ihrke, M., Herrmann, J. M., & Hasselhorn, M. (2010). Event-Related Brain Potential Correlates of Identity Negative Priming from Overlapping Pictures. *Psychophysiology*, *47*(5), 921-930. https://doi.org/10.1111/j.1469-8986.2010.00989.x

(10) Ritschel, T., Ihrke, M., Frisvad, J. R., Coppens, J., Myszkowski, K., & Seidel, H. (2009). Temporal Glare: Real-Time Dynamic Simulation of the Scattering in the Human Eye. *Computer Graphics Forum*, *28*(2), 183-192. https://doi.org/10.1111/j.1467-8659.2009.01357.x

(9) Ritschel, T., Smith, K., Ihrke, M., Grosch, T., Myszkowski, K., & Seidel, H. (2008). 3D Unsharp Masking for Scene Coherent Enhancement. *ACM Transactions on Graphics*, *27*(3), 90:1-90:8. https://doi.org/10.1145/1399504.1360689

(8) Schrobsdorff, H., Ihrke, M., Kabisch, B., Behrendt, J., Hasselhorn, M., & Herrmann, J. M. (2007). A Computational Approach to Negative Priming. *Connection Science*, *19*(3), 203-221. https://doi.org/10.1080/09540090701507823

Book chapters

(7) Csifcsák, G., Forstmann, B., & *Mittner, M.* (2021). Transcranial stimulation and decision-making. In Wassermann, E. M., Peterchev, A. V., Ziemann, U., Lisanby, S. H., Siebner, H. R., & Walsh, V. (Ed.), *The Oxford Handbook of Transcranial Stimulation*. https://www.oxfordhandbooks.com/view/10.1093/oxfordhb/9780198832256.001.0001/oxfordhb-9780198832256

(6) Ihrke, M., Schrobsdorff, H., & Herrmann, J. M. (2009). Denoising and Averaging Techniques for Electrophysiological Data. In Velazquez, J. L. P. & Wennberg, R. (Ed.), *Coordinated Activity in the Brain: Measurements and Relevance to Brain Function and Behavior* (pp. 165-189). Springer New York. https://doi.org/10.1007/978-0-387-93797-7_9

Conference proceedings

(5) Schrobsdorff, H., Ihrke, M., & Herrmann, J. M. (2013). Modeling Structure and Dynamics of Selective Attention. In *Biologically Inspired Cognitive Architectures 2012* (pp. 287-295). https://doi.org/10.1007/978-3-642-34274-5_50

(4) Ihrke, M., Ritschel, T., Smith, K., Grosch, T., Myszkowski, K., & Seidel, H. (2009). A perceptual evaluation of 3D unsharp masking. In *Human Vision and Electronic Imaging XIV* (pp. 72400R). https://doi.org/10.1117/12.809026

(3) Ihrke, M., Schrobsdorff, H., & Herrmann, J. M. (2009). Recurrence-Based Synchronization of Single Trials for EEG-Data Analysis. In *Intelligent Data Engineering and Automated Learning - IDEAL 2009* (pp. 118-125). https://doi.org/10.1007/978-3-642-04394-9_15

(2) Ihrke, M., Schrobsdorff, H., & Herrmann, J. M. (2008). Compensation for Speed-of-Processing Effects in EEG-Data Analysis. In *Intelligent Data Engineering and Automated Learning – IDEAL 2008* (pp. 354-361). https://doi.org/10.1007/978-3-540-88906-9_45

(1) Yoshida, A., Ihrke, M., Mantiuk, R., & Seidel, H. (2008). Brightness of the Glare Illusion. In *Proceedings of the 5th Symposium on Applied Perception in Graphics and Visualization* (pp. 83-90). https://doi.org/10.1145/1394281.1394297

Grants and Scholarships

Research Grants

UiT Talent Infrastructure (4.6 million NOK, ≈ 420,000 EUR) — 2025

- COCAMAP: Cognitive causal mapping: Stimulation-based brain-mapping of cognitive functions
- acquisition of a robotic system for cognitive causal mapping

MSCA-PF (host) (2.4 million NOK, ≈ 210,000 EUR) — 2024/25 - 2026/27

- Controlling the flow of attention: Transition dynamics of attention and distraction across external and internal domains (MIND-FLOW)
- 2 year post-doc project

UiT High Score/UiT Talent (3 million NOK, ≈ 350,000 EUR) — 2024 - 2027

- Neural mechanisms of off-focus cognition
- 3 year full-time post-doc project

HelseNord project PFP1237_15 (3 million NOK (≈ 350,000 EUR)) — 2015 - 2018

- Optimizing transcranial direct current stimulation for treatment of chronic pain and depression
- 3 year full-time post-doc project

PhD candidate from Health faculty, UiT — 2016 - 2020

- Neural and computational correlates of mind wandering
- 4 year PhD project

Personal Scholarships and Awards

Incentive grant (200k NOK, ≈ 18,000 EUR) — 2017

- Grant awarded by the Faculty of Health for excellent grades in external applications

Incentive grant (200k NOK, ≈ 18,000 EUR) — 2016

- Grant awarded by the Faculty of Health for excellent grades in external applications

Research Prize — 2015

- Research prize from the Institute for Psychology

GGNB Fellowship — 2009/05 - 2012/01

- Full-time PhD Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

GGNB Bridging Fund Scholarship — 2008/10 - 2008/12

- Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

DFG Scholarship — 2009/01 - 2009/04

- Full-time Scholarship "Passungsverhältnisse schulischen Lernens: Verstehen und Optimieren"

Erasmus Scholarship — 2004/07 - 2005/08

- Scholarship for studies abroad

Presentations

Invited Presentations

Cognitive Neuroscience Meeting (CNS) — Boston, US

Selected invited presentations — 2025

Seminar series Max-Planck Institute for Human Development — Berlin, Germany

Invited seminar 2025

IMBIT seminar series, University of Freiburg — Freiburg, Germany

Invited seminar 2025

Institute du Cerveau — Paris, France

Invited workshop lecturer 2024

Network Modeling for Transcranial Magnetic Stimulation (NEMO-TMS) — Freiburg, Germany

Invited presentation 2023

University of Oslo — Oslo, Norway

Invited symposium 2023

International Conference on Cognitive Neuroscience (ICON) — Helsinki, Finland

Invited presentation 2022

Laboratoire de Sciences Cognitives et Psycholinguistique, École Normale Supérieure — Paris, France

Invited presentation 2022

Organization for Human Brain Mapping (OHBM) — Online

Invited presentation 2021

NRSN National Neuroscience Symposium — Norway

Invited presentation 2021

fMRI 20-year Anniversary Seminar, University of Trondheim — Trondheim, Norway

Invited presentation 2019

The Role of Neural Oscillations in Human Cognition (Summer School) — Uslar, Germany

Invited presentation 2017

Mind-Wandering Symposium, University of Amsterdam — Amsterdam, The Netherlands

Invited presentation 2016

International Conference on Cognitive Neuroscience (ICON) — Brisbane, Australia

Invited presentation 2014

Supervision

Post-doctoral researchers

Vesal Rasoulzadeh	"Time-on-task effects on attentional transitions" — *visiting post-doc (Roeljan Wiersema)*	2026
Sam Verschooren	"Controlling the flow of attention: Transition dynamics of attention and distraction across external and internal domains"	2025
Josephine Groot	"Neural mechanisms of off-focus cognition"	2024
Hema Nawani	"ADHD and mind-wandering" (co-supervisor, main supervisor: Gábor Csifcsák)	2020-2023
Gábor Csifcsák	"Optimizing transcranial direct current stimulation for treatment of chronic pain and depression" — *HelseNord, PFP1237_15*	2016-2019

PhD students

Espen Wiseman Engeland	"Individual Variability in early treatment response to intermittent Theta-Burst Stimulation (iTBS) in Depression: Results from a Randomized, Double-Blind, Sham-Controlled Study" (co-supervisor, main supervisor: Marte Ørbo)	2025-2029
Steffen Rygg Aasen	"Executive functions and mind wandering" (co-supervisor, main supervisor: Gábor Csifcsák)	2024
Andreas Alexandersen	"Unraveling the cognitive and neural mechanisms of mind wandering using transcranial electric stimulation"	2023
Maren Angel Christensen	"Structural differences in brain structure after trauma: An analysis based on the HUNT study and the Tromsø Study"	2022
Malin Gullsvåg	"The relation between respiratory mechanisms and cognitive abilities in adolescence and adulthood" (co-supervisor, main supervisor: Claudia Rodríguez-Aranda)	2020
Josephine Groot	"Neural mechanisms of off-focus cognition"	2018-2023
Nya Mehnwolo Boayue	"Neural and computational correlates of mind wandering"	2016-2020
Isabel Viola Kreis	"Too precise or too imprecise: which parameter is gone awry in autism and psychosis" (co-supervisor, main supervisor: Gerit Pfuhl)	2017-2021

Zsolt Turi	"Methodological and cognitive aspects of transcranial electrical stimulation" (co-supervisor, main supervisor: Andrea Antal)	2011-2015
Wouter Boekel	"When the brain takes a break: Neural correlates of mind-wandering" (co-supervisor, main supervisor: Birte Forstmann)	2013-2017

Research Line students (forskerlinje)

Almina Selimovic	"Through the lens of attention: Factors that systematically modulate pupil dilation" (co-supervisor, main supervisor: Josephine Groot)	2025
Mathilde Aursand	"Being with the body: The impact of meditation on interoceptive attention" (co-supervisor, main supervisor: Sam Verschooren)	2025
Ragnhild Drevland	"The effect of cTBS over the left AG on mind wandering" https://osf.io/vjsm7	2021-2024
Steffen Rygg Aasen	"The effect of iTBS over the left DLPFC on mind wandering" https://osf.io/txpu2	2021-2024
Samy Babiker	"Modulating loss of control using rTMS during chronic pain" (co-supervisor, main supervisor: Gábor Csifcsák) https://osf.io/r74uk	2021-2024
Andreas Alexandersen	"Investigating the relationship between executive control and mind-wandering using transcranial direct current stimulation"	2020-2022

Master students

Maria Puchinina	"Immediate TMS-evoked potentials: A measure of cortical excitability?"	2026-2027
Anastasios Ntousas	"Effects of motivation on reinforcement learning and mind wandering" (co-supervisor, main supervisor: Gábor Csifcsák)	2026
Tim Cools	"Does mind wandering modulate the relationship between gaze and stimulus value?" https://osf.io/ca95r	2023-2024
Kjell Slupphaug	"Moderation and interaction effects in structural equation modeling" (co-supervisor, main supervisor: Mehmet Mehmetoglu) https://modsem.org/	2023-2024
Ceylin Karayel	"Exploring the Link Between Mind-Wandering and Reinforcement Learning" https://osf.io/ca95r	2023-2024
Krister Karlsen	"Mind wandering and online experimentation" https://osf.io/wjvk2	2021-2022
Maren Christensen	"Volumetric changes in brain structures due to trauma"	2021-2022
Andreas Alexandersen	"The effect of transcranial direct current stimulation on the interplay between executive control, behavioral variability and mind wandering: A registered report" https://hdl.handle.net/10037/25274	2020-2022
Caroline Angen **Ina Klakegg** **Anastasija Kuprejeva**	"The effect of chronic pain and loss of control on value-based decision making" (co-supervisor, main supervisor: Gábor Csifcsák)	2020-2021
Eirik Melsæter	"Intermittent absence of control during reinforcement learning interferes with Pavlovian bias in action selection" (co-supervisor, main supervisor: Gábor Csifcsák) https://osf.io/89mdr	2019
Tarjej Hotvedt	"Mindfulness and Psoriasis: A pilot study" https://munin.uit.no/handle/10037/15507	2016
Are Svendsen	"Eksistensiell dynamisk terapi på inneliggende pasienter - en eksplorerende studie av utfall og relasjonelle virkningsmekanismer" (co-supervisor, main supervisor: Jørgen Sundby)	2014-2015
Ida Marie Opdahl	"Wandering body, wandering mind? The relationship between bodily movement, creativity and mind-wandering" https://munin.uit.no/handle/10037/9143	2014-2015
Ira Beckmann	"Negatives Priming als Identifikations- oder Selektionsphänomen" (co-supervisor, main supervisor: Jörg Behrendt)	2010

Professional students (Hovedoppgave)

Ida Jespersen Rønningsen **Marcus Buvik**	"The effect of excitatory and inhibitory TBS over the left AG on mind wandering" https://osf.io/x2vmt	2023
Jostein Ilvær **Andreas Thommesen**	"The effect of transcranial temporal interference stimulation (tTIS) on value-based decision making in healthy adults"	2022

Emilia Judeth Olsen **Nora N. Holmboe**	"Parafilia og selvregulering: En normeringsstudie" https://osf.io/sgk8v	2022
Robert Taknes **Tim Lockertsen**	"Psykologi i rettens tjeneste: Hvilke faktorer påvirker årsaksattribusjon i erstatningssaker for psykisk helseskade?" (co-supervisor, main supervisor: Jørgen Sundby) https://munin.uit.no/handle/10037/21126	2020
Pål Ovanger Stensland	"The role of eye-movements and pupillometry in the representation of value in a reinforcement-learning context"	2021
Celina Jakobsen **Joakim**	"Klinikers vurdering mtp kjønn og ADHD utredninger" (co-supervisor, main supervisor: Jørgen Sundby)	2021

Bachelor students

Vetle Bersås **Zuzanna Kilian**	"Effects of motivation on reinforcement learning and mind wandering" (co-supervisor, main supervisor: Gábor Csifcsák)	2026
Christian Fossheim	"Moment-to-moment fluctuations in emotional valence and mind-wandering"	2015
Ingrid Marie Skjerstad	"The wandering mind of the long-distance runner"	2017
Anna Elfrida Vollsund **Iselin Finn**	"Effectiveness of HD-tDCS on influencing executive control during mind wandering" https://osf.io/nm2sz/	2018
Kristian Østhagen Haukås	"The wandering mind of the long-distance runner"	2019
Aurora Vangen **Samy Babiker** **Sanna Sandell** **Silje Haugan** **Steffen Rygg Aasen**	"The effect of sleep-deprivation on mind-wandering: an EEG study" (co-supervisor, main supervisor: Gábor Csifcsák) https://osf.io/xq6wr	2021
Emily Haga **August Lullau**	"Transcranial Magnetic Stimulation Efficacy in Depression - A Systematic Review and Meta-Analysis"	2021
Caroline Angen **Marlene Holdt**	"Employment of mental effort and representation of environmental volatility during intermittent absence of control over rewards and losses under different feedback schedules" (co-supervisor, main supervisor: Gábor Csifcsák) https://osf.io/7wcej/	2020
Jorunn Bjørkøy **Sarjo Kuyateh** **Håkon Reithe**	"Transcranial direct current stimulation above the medial prefrontal cortex facilitates decision-making following periods of low outcome controllability" (co-supervisor, main supervisor: Gábor Csifcsák) https://psyarxiv.com/rbeuz/	2018

Exchange students and Interns

Stefan Weinhardt	"The effect of respiration on internal and externally directed attention" (co-supervisor, main supervisor: Sam Verschooren) https://osf.io/xgjyr	2025
Thea Waldleben	"High-precision multi-session estimation of individual functional connectivity networks"	2025
Federica Luzzi	"The effect of transcranial temporal interference stimulation (tTIS) on value-based decision making" https://osf.io/rc756	2022
Sven Wientjes	"Hidden Markov Models for estimating attentional focus"	2020
Terezie Sedlinska	"The impact of loss-of-control on system arbitration" https://osf.io/73huk	2019-2021
Tabea Brödel	"Methods for blink detection in pupillometric data"	2021
Isabel Kreis	"The role of executive functions in mind wandering" https://doi.org/10.1111/ejn.15049	2016
Carole Schmidt	"The role of executive functions in mind wandering" https://doi.org/10.1111/ejn.15049	2017
Divyaratan Popli	"Eye-tracking and pupillometric correlates of mind wandering"	2013

Teaching

Lectures and supervision (University of Tromsø)

Tromsø, Norway

from 2014

Undergraduate

- Methods and Statistics III (PSY-1513)
- Methods and Statistics I (PSY-1511)

- Open Science (PSY-2901)
- Cognitive Neuroscience (PSY-2547, PSY-2553)
- Affective Neuroscience (PSY-2549)
- Biological Psychology (PSY-1004)
- Psychometrics (PSY-1504)
- Practical Course: Gerontopsychology (PSY-2543)
- Research Practical (PSY-1509/1510)
- Research skills for Master students (PSY-3900)
- Research communication (PSY-1702)

PhD Courses

- Quantitative Research Methods (HEL-8024)
- Applied Linear Regression (HEL-8030)
- Research Design and Statistics (PSY-8002)
- Quantitative Method in Clinical Health Research (HEL-8017)
- Informal reading club for PhD students and faculty: Bayesian statistics

Lectures and supervision (NTNU, Trondheim) — Trondheim, Norway

from 2021

Continuing Education and Professional Development

- Learning R for research and reporting: From basics to advanced applications (PSY6016)

PhD Courses

- Multivariate quantitative research methods (PSY8003)

Undergraduate

- Research Methods (PSY3100)
- Statistics and quantitative research methods (PSY2017)
- Research Design (PSY2022)

Lectures and supervision (University of Amsterdam) — Amsterdam, The Netherlands

summer 2013

Master

- Introduction to Neuroscientific Methods and Brain Anatomy: lecture and practical on electroencephalography

Lectures and supervision (University of Göttingen) — Göttingen, Germany

GGNB Methods Course (introductory level)

- Introduction to Python Programming (A123) – winter term, 2011/2012
- Matlab and Python Programming (A109) – winter term, 2010/2011

Teaching Seminar

- Gestaltung von Lehr-Lernprozessen – summer term, 2010: Introductory seminar to the psychology of learning and teaching for students pursuing teaching certification

Service and Leadership

Appointments

Head of research, Institute of Psychology, UiT - The arctic university of Norway, Norway — 2022--2024

Research group leader (RG Cognitive Neuroscience), Institute of Psychology, UiT - The arctic university of Norway, Norway — 2022--

Selected advisorial and managerial activities

Management committee COST action CA24148 — 2025
https://www.cost.eu/actions/CA24148/

Reviewer Riksbankens Jubileumsfond RJ — 2025

Review panel member, online course on 'Statistical research methods', Epigum (Sage Publications) — 2024

European Research Council (reviewer for ERC Consolidator Grant, ECoG) — 2024

Reviewer The French National Centre for Scientific Research (CNRS) - ATIP - Avenir program — 2022

Reviewer Netherlands Organisation for Scientific Research (NWO) - Vici Social Sciences and Humanities — 2019

Advisor Hertie Foundation Project: Improvement of school education by neuroscientific research 2017

Professional development

Research group leader development program (University of Tromsø) 2021--2022

Program for young research leaders; joint program University of Oslo, University of Bergen, University of Trondheim, University of Tromsø - leadership, networking, financing, communication, strategy 2014--2015
https://www.uib.no/psyfa/43678/program-yngre-forskningsledere

Recording and Analysis of Eye-Movements, Cologne, Germany 2008

Computational Neuroscience at the Max-Planck-Institute for Dynamics and Self-Organization 2005

Institutional Responsibilities

Selection committee for Associate professor in quantitative methods (University of Trondheim) 2022

Evaluation committee for evaluation of qualification for professorship (University of Tromsø) 2022

Head of the Research Ethics Committee at the Institute for Psychology (IPS-REC, University of Tromsø) 2022--2024
https://en.uit.no/research/ethics/ips

Head of committee IPS Research forum (University of Tromsø) 2022--2024

Member of faculty-level PhD committee (University of Tromsø) 2022--2024

Head of selection committee for Associate professor in cognitive neuroscience (University of Bergen) 2021

Selection committee for PhD positions (University of Trondheim, University of Bergen, University of Helsinki) 2021

Evaluation committee for several PhD and post-doc positions (University of Tromsø) 2014--

Skills

Experimental Methods
fMRI, EEG/ERP, EOG, tDCS, TMS, eye-tracking, pupillometry

Programming Languages
Very Good: Python, R, C, C++, Matlab/Octave
Good: Perl, Java, SQL, PHP, Presentation
Moderate: F77, Assembler, Objective C, Bash, Javascript

Computing
Parallel Programming (MPI), GPU Programming (OpenGL), Unix/Linux Administration, LaTeX, HTML/CSS

Languages

German (native)
English (fluently written, read and spoken)
Norwegian (fluently written, read and spoken)
French (basic skills)
Dutch (basic skills)

Interests

running, skiing, judo, rock-climbing, hiking, table-tennis, football